



Quantum recurrent neural network for sequential labeling

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Abstract

Quantum computing is a computing paradigm that follows the laws of quantum mechanics, enabling the efficient solution of complex computational problems beyond the capabilities of classical systems. A combination of quantum computing and machine learning, known as quantum machine learning (QML), is a rapidly emerging field that has the potential to revolutionize traditional artificial intelligence workflows. Recently, QML algorithms have been applied to various natural language processing (NLP) applications, yielding promising results. However, little attention has been devoted to applying quantum-enhanced models to sequential labeling, a crucial task in NLP. To bridge this gap, this paper proposes a quantum recurrent neural network (QRNN) designed specifically for sequential labeling tasks. The proposed QRNN is a hybrid quantum-classical model, which combines quantum and classical resources. It utilizes data angle encoding as encoding methods to convert classical data into quantum states and variational layers to process quantum data. Comprehensive experiments are conducted on nine distinct code-mixed POS tagging datasets, covering multiple language pairs, to evaluate the performance and generalization of the QRNN. Additionally, the study includes ablation analyses that investigate the impact of various data encoding schemes, entanglement strategies, and depths of the variational layer on model performance. The experimental results demonstrate that the proposed QRNN consistently achieves higher tagging accuracy and fewer trainable parameters compared to both classical RNN models and state-of-the-art quantum long short-term memory (QLSTM).

Keywords Natural language processing · Machine learning · Quantum computing · Quantum machine learning · Quantum neural network

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1 Introduction

Machine learning is a branch of artificial intelligence that makes computer systems learn patterns and make predictions from data without explicit programming. Its potential to adapt and improve through experience makes it an essential methodology in various domains. One of the most well-known applications of machine learning is natural language processing (NLP), where machine learning models are utilized to analyze and process text data in such a way that systems can understand linguistic patterns, semantic relationships, and contextual meaning [1]. Moreover, the introduction of neural networks has revolutionized NLP, especially with the advent of recurrent neural networks [2], which effectively handle sequential data, such as sentences and speech. Recently, transformer-based models [3] have outperformed state-of-the-art models in performing several NLP tasks, due to their ability to handle long-range dependencies.

However, the superiority of transformer-based models poses various challenges, such as demands for a large amount of data and parameters to train models [4]. For example, large language models require trillions of tokens and billions of parameters during training [5]. Training such models requires considerable computational power, time, and hardware resources, making it feasible for only a few organizations. Therefore, there is a demand for a new computing methodology that can enhance computational efficiency and lead to further advancements.

An alternative computing methodology is quantum computing [6], which utilizes the laws of quantum mechanics to improve computational performance and information processing. Quantum computing can efficiently solve problems that are difficult for traditional computing, such as factoring large numbers [7]. Recently, quantum machine learning (QML) [8], a subfield of quantum computing that integrates machine learning methodology, has received significant attention. QML leverages the principle of quantum mechanics to accelerate machine learning tasks or establish new expressive models [9]. However, current quantum computers refer to noisy intermediate-scale quantum (NISQ) systems [10] that are specified as noisy and unstable [11]. To effectively utilize current quantum devices, hybrid classical-quantum algorithms [12] have been introduced, using variational quantum circuits (VQCs). VQCs employ trainable parameters and can learn task-specific representations, offering greater expressiveness and adaptability compared to classical models [13].

Hybrid QML models have shown promising performance in various domains, including optimization [14], pattern recognition [15], and recently natural language processing tasks [16]. Although several studies have explored the application of QML techniques to NLP such as language modeling [17], text classification [18], and sentiment analysis [19], there is still a noticeable research gap in applying quantum-enhanced architectures to sequential labeling, a core problem in NLP that involves assigning categorical labels to each token in a sequence. However, hybrid QML models can take advantage of contextuality to represent complex sequence relationships compared to classical sequential models [20].

To overcome this research gap, we propose a hybrid quantum-classical recurrent neural network (QRNN) that is used specifically for sequential labeling tasks. The proposed architecture utilizes the expressive power of VQCs as recurrent layers to capture temporal dependencies within text sequences, while the classical layers are employed to perform the final classification of text data into the corresponding labels. Our main contribution is: 1. to develop a QRNN that combines VQCs with classical layers to perform efficient sequential labeling; 2. to evaluate QRNN performance employing part-of-speech (POS) tagging, a sequence labeling task on nine different code-mixed datasets, and 3. to present ablation stud-

ies that examine the influence of different data encoding methods, entanglement strategies, and variational circuit depths on model performance. The experimental result shows that the proposed QRNN achieves higher accuracy with fewer trainable parameters compared to classical RNN and state-of-the-art quantum long short-term memory [21] (QLSTM).

The remainder of this paper is as follows. Section 2 discusses the essential background of quantum computing, while Sect. 3 explores a related work on QNNs. Section 4 discusses the basic structure of a recurrent neural network, followed by Sect. 5, which explains the architecture of the proposed QRNN model. Section 6 describes the dataset applied for the experiment, and Sect. 7 details the experiment and result of our proposed model. Finally, Sect. 8 concludes the paper with a conclusion.

2 Background

This section provides an overview of the preliminary concepts of quantum computing, such as quantum information and properties, quantum operators and circuits, and VQC.

2.1 Quantum bit (Qubit)

A qubit is an essential quantum computing processing element analogous to the classical bit. In contrast to a classical bit that can be 0 or 1, a qubit can be 0, 1, or a linear combination of both states, known as the superposition state. Usually, the Dirac notation describes the primary states of a qubit: $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

2.2 Superposition & Entanglement

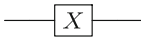
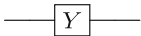
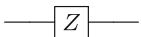
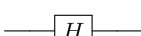
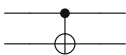
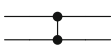
Quantum computing follows the principles of quantum mechanics when performing computation. Superposition is one of the crucial principles of quantum mechanics that allows qubits to exist in a linear combination of all possible values simultaneously. For example, the superposition state of the single-qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are respective amplitudes of state 0 and 1. Similarly, superposition state of two qubits: $|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$. In general, the superposition of any n-qubit system can combine 2^n possible states simultaneously. This property allows for the generation of exponential computational states with n bits and the processing of them in parallel [22].

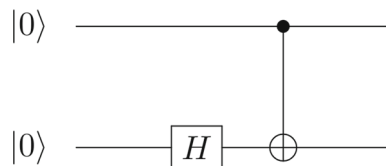
Entanglement is another essential principle of quantum mechanics that defines correlations among multiple qubits. When qubits are entangled, the states of one qubit become intrinsically linked to another quantum qubit. Moreover, measuring one entangled qubit state can immediately affect the impact on the other entangled qubit state. This property makes quantum computing execute information extremely interconnected and correlated, offering more powerful and efficient computations [22].

2.3 Quantum gate

Quantum gates are fundamental operators that perform computations on quantum information. They are similar to classical computing gates, which apply Boolean operations to classical bits. Meanwhile, quantum gates perform linear operations on qubits to manipulate

Table 1 Examples of single- and multiple-qubit gates commonly used in quantum algorithms

Gate name	Unitary matrix representation	Diagrammatic representation
Pauli-X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	
Pauli-Y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	
Pauli-Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	
Hadamard	$\begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & -\frac{1}{\sqrt{2}} \end{bmatrix}$	
Controlled Not (CNOT)	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	
Controlled-Z	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$	

**Fig. 1** A quantum circuit generating one of Bell's maximally entangled states. This circuit begins with the initial state as $|00\rangle$ and applies quantum operator as Hadamard and CNOT

their quantum state. Mathematically, unitary matrices represent quantum operators that apply one quantum state and transform it into another valid one, as shown in Eq. 1. The definition of the unitary matrix ($U^{-1} = U^\dagger$), where U represents the unitary matrix of a quantum gate and $U^\dagger$ is adjoint to U , ensures that quantum gates are reversible operations. Table 1 presents the frequently used quantum gates to perform specific tasks in quantum algorithms.

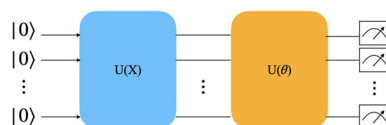
$$U|\psi\rangle = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix} = |\phi\rangle \quad (1)$$

Quantum circuits are the fundamental building blocks of quantum information. Quantum circuits, consisting of a series of quantum gates, are essential elements in quantum algorithms, integrating qubits to represent quantum information, wires to flow information, and quantum gates to perform a specific task, as illustrated in Fig. 1. This example contains a Hadamard gate followed by a CNOT gate that generates maximally entangled states.

2.4 Variational quantum circuit (VQC)

VQCs [23] are a specific type of quantum circuit consisting of adjustable parameters optimized through an iterative process. Figure 2 describes the typical structure of VQC, which

Fig. 2 A general architecture of VQC that includes $U(X)$ and $U(\theta)$ unitary operators encode classical data and process the data



includes three layers: a data encoding layer, a variational layer, and a measurement layer. The data encoding layer transforms classical data into quantum data from the initial state $|0\rangle^n$ to a target state by the $U(X)$ operator. The variational layer contains the tunable parameters and applies operations with $U(\theta)$ to process the quantum data, and ultimately, the measurement gives the computational results of the VQCs.

3 Related work

The development of the HHL algorithm [24] to solve linear equations on quantum computers marked the beginning of QML algorithms. HHL algorithms are applied in quantum support vector machine [25] and quantum principle component analysis [26] to perform linear algebra computation that shows exponential speedup compared to their classical versions. However, these algorithms cannot be applied to current quantum hardware due to their dependence on quantum RAM to execute.

In this NISQ era, the primary focus is to develop hybrid quantum algorithms that utilize VQCs in a short depth and embedded in quantum and classical resources by the variational loop. In particular, variational quantum algorithms (VQAs) [27] become central to quantum learning algorithms of the NISQ era, for example, the variational quantum eigensolver (VQE) [28], which expressive and trainable structures have laid the foundation for quantum neural networks (QNNs) [29]. Therefore, hybrid algorithms have become an integral part of research in quantum algorithms for machine learning, neural networks, and optimization problems.

Several studies have explored the development of hybrid QNNs to apply to real-world problems, and they demonstrate improved accuracy compared to their classical counterparts. For example, Liu et al. [30] have developed a quantum convolutional neural network (QCNN) that combines a quantum-classical paradigm to improve the feature mapping procedure. QCNN can run on NISQ quantum devices and perform better classification tasks than classical CNN. Cheng et al. [31] proposed that QCNN model uses parameterized quantum circuits to replace the convolutional layer with the quantum convolutional layer to enhance the extraction capabilities of hidden information from images. It is applied to the MNIST dataset to perform binary classification, showing a higher performance than its classical counterpart.

Moreover, some studies implement quantum recurrent neural networks (QRNN), such as Ceschini et al. [32], who developed a hybrid QRNN to forecast time series data for renewable energy. QRNN replaces the classical layer with a variational layer and shows improved results compared to its classical version in the prediction error. In addition, Chen et al. [33] implemented quantum long- and short-term memory (QLSTM), which can be executed on current quantum devices for sequential modeling. QLSTM is used to evaluate the function values in various time instances. In contrast, the performance of their proposed model on non-trivial sequential data needs to be verified.

Researchers are now trying to implement the QRNN model for NLP applications. For example, Sipio [34] implements a hybrid QLSTM model and utilizes it to implement POS tagging, which initiates an attempt to perform NLP applications using quantum computing. However, this QLSTM, a basic model for NLP applications, can handle only small datasets.

Pandey et al. [35] also implemented QLSTM to perform POS tagging for the Mizo language, a low-resource Indian language. However, this model can only perform better on a small dataset. Pandey et al. [21] proposed a QLSTM model that replaced LSTM gates with VQCs and applied it to POS tagging on code-mixed social media datasets. Their experimental result shows a significant improvement over the classical LSTM with the same structure. Yu et al. [19] introduced a QRNN model that integrates a pre-trained BERT model for word embedding. This QRNN was applied to perform sentiment analysis in Bengali languages, achieving results that exceeded classical models.

Quantum natural language processing (QNLP) [36] is another framework for implementing NLP applications on near-term quantum computers. QNLP combines both the NLP compositional structure and distributional semantics paradigms as a DiscoCat module. It represents the grammatical structure of sentences in Pregroup grammar and words of sentences in string diagrams. Then, DiscoCat transforms these string diagrams into quantum circuits for further processing. Several NLP applications have utilized the QNLP framework, such as sentiment analysis [37], question answering [38], and grammar-sensitive classification [39]. However, QNLP requires vast resources for computations, making it time-consuming to process the data.

The most existing QRNN performs focused on high-level NLP application such as sentiment analysis, text classification, and forecasting the time series data. However, QLSTM [21] shows the feasibility of performing the POS tagging task. However, it only works on small datasets, which makes it unsuitable for large datasets. Furthermore, the current QRNN model follows fixed data encoding and entanglement strategies, which restrict their flexibility and generalization capability. To overcome these gaps, the proposed QRNN is well suited for large datasets for sequence labeling. In addition, we performed a several experiments to demonstrate the effect of data encoding, entanglement strategies, and depth of variable layer on model's performance.

4 Recurrent neural network

Recurrent neural networks (RNNs) are a popular variant designed to process sequential data of varying lengths. It is especially effective for tasks with sequential dependencies, including text data, speech recognition, and time series analysis. The critical feature of RNN is its ability to retrieve data from the previous state to predict the present state. This process utilizes the recurrent connections that allow the flow of information from one state to another.

Figure 3 shows the structure of an RNN cell, a basic composition unit for RNN models. Each RNN cell has a hidden state, a data instance, and a prediction state. The stacking of multiple cells forms RNN models, as shown in Fig. 4. Each cell processes a single data instance and passes the information to the next cell.

RNN takes input in each time step to process sequential data and updates its hidden state according to previous hidden states and current input, as given in Eq. 3. The hidden state behaves as memory, which holds the past information and contributes to handling time series data. These characteristics of RNN allow for the flow of data from one time step to the next time step. Finally, the prediction of the model depends on the current hidden state, as represented in Eq. 4.

$$h_t = f(U * x_t + V * h_{t-1}) \quad (2)$$

$$y_t = g(W * h_t) \quad (3)$$

Fig. 3 The architecture of an RNN cell that utilizes U , V , and W which are weight matrices to learn sequential data

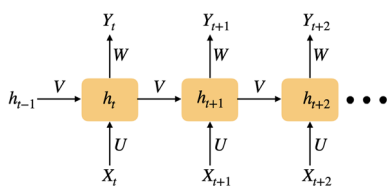
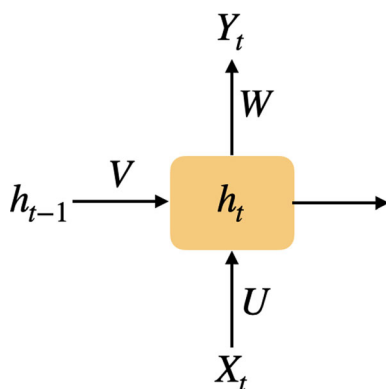
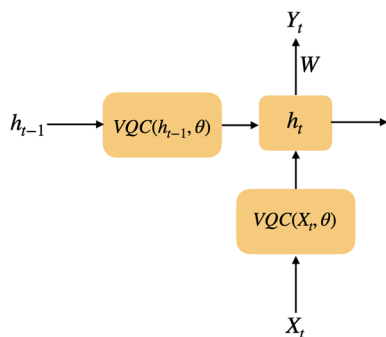


Fig. 4 The typical structure of RNN models is composed by stacking different RNN cells, where X_t , X_{t+1} represent data instances at different time steps, and h_{t-1} , h_t , h_{t+1} are hidden instance that capture sequential dependencies. The outcome of each cell is reflected by Y_t , Y_{t+1} , Y_{t+2} of different time instances

Fig. 5 The structure of a QRNN cell processes a data instance and a hidden state through VQCs while generating outputs from a classical layer



5 Quantum recurrent neural network

This study introduces a quantum-classical hybrid QRNN that combines quantum and classical methodology to perform sequential labeling. In each cell of the proposed QRNN, the hidden and input layer is replaced with VQCs, while the output layer is classical, as shown in Fig. 5. Implementing VQC includes data encoding, variational layers, and measurement layers. The data encoding module encodes classical information into quantum information, and the variational layers process quantum information through learnable parameters. Finally, the measurement layer provides the classical outcome of processed data.

5.1 Data encoding

Quantum computers can only understand quantum information, making it necessary to encode classical data into quantum states. This encoding process, known as encoding methods or data embedding, plays an essential role in designing QML algorithms, as it maps classical information into quantum-supportable information formats. The selection of an encoding method significantly affects the performance of QML algorithms [40]. Common approaches to data embedding methods in quantum algorithms are basis, amplitude, and angle encoding.

Basis encoding [41] associates the basis states of an n -qubit system with bit strings of the binary representation of input features. The M -bit string of the input map corresponds to the M -qubit quantum system. Suppose that a dataset X has n samples and each has m characteristics. Using base encoding to encode the dataset, the bit string m of a sample $x^i = (b_1, \dots, b_m)$ corresponds to the corresponding m -qubit system $|x^i\rangle = |b_1 \dots b_m\rangle$, where $b_m \in \{0, 1\}$ and $i \in \{1, \dots, n\}$.

$$|X\rangle = \sum_{i=1}^N |x_i\rangle \quad (4)$$

The amplitude encoding [42] normalizes the classical data X , $\sum_i |x_i|^2 = 1$ and maps into the amplitudes of the quantum states. It is an efficient encoding technique that consumes fewer qubits to encode higher-dimensional data. Suppose a dataset X consists of n samples with m features; amplitude encoding requires $N \geq \log nm$ qubits to embed the entire dataset.

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_{2^n} \end{bmatrix} \leftrightarrow |\psi\rangle = \sum_{i=1}^{i=2^n} x_i |i\rangle \quad (5)$$

However, amplitude encoding converts entire data into quantum states at a time that requires many quantum gates. In contrast, angle encoding [43] utilizes quantum rotational gates to represent classical data in quantum states. Angle encoding operates at individual data points at a time, which makes it efficient in processing time evolution data. For example, the characteristics $N \{x_1, \dots, x_n\}$ of the data point can be represented by the angle encoding method using the following equation.

$$|X\rangle = \bigotimes_{i=1}^N \cos x_i |0\rangle + \sin x_i |1\rangle \quad (6)$$

QRNN employs the angle encoding method for data encoding, which is especially well suited for QNN models. Figure 6 illustrates that the quantum circuits map each data point $(x_1, x_2, \dots, x_n)$ to the quantum states using rotation gates R_x . We aim to choose this encoding that should be simple and efficient for mapping classical data to quantum data. This encoding requires a few quantum gates to encode the data and encode the data at each time instance, which is necessary for RNN to process the sequential data efficiently.

In angle encoding, each qubit can only represent a single feature value, which limits QRNN to processing high-dimensional data. To overcome this issue, we added an autoencoder at both the input and the hidden layer. The autoencoder compresses high-dimensional vectors into a lower-dimensional latent representation while preserving the essential information. Although

Fig. 6 A quantum circuit that QRNN utilizes to encode classical data in quantum states

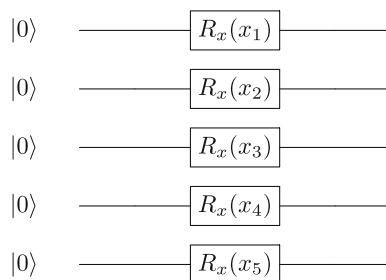
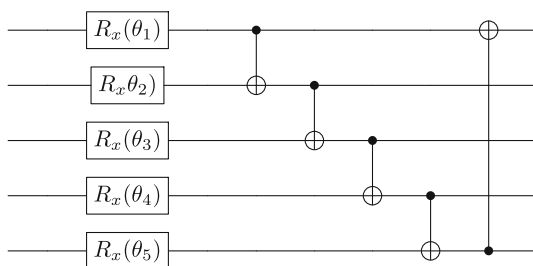


Fig. 7 A quantum circuit of a variational layer of the proposed VQC



angle encoding is used in the primary architecture, we also evaluate amplitude encoding as an alternative data encoding strategy. Experiments analyzing the impact of different encoding methods on QRNN performance are presented in Subsect. 7.3.

5.2 Variational layer

Generally, variational layers map the encoded data to approximate the target function using adjustable parameters, similar to traditional neural networks operating in approximation theory [44]. In VQCs, variational layers have fixed structures, including parameterized quantum gates and entanglement layers between two qubits [45]. The parameterized quantum gate consists of the tunable parameters initialized randomly to train the QNN models through optimization. The entangling layer generates correlated quantum states between qubits. Moreover, different entanglement strategies provide different impacts on performance of the proposed model. The detail explanation is provided about various entanglement approaches in variational in Subsect. 7.4.

Figure 7 represents the variational layer circuit for our proposed VQC that consists of single-qubit rotation $R_x(\theta_i)$ for each qubit followed by a chain of CNOT gates. The single-qubit rotation gates include adjustable parameters like $(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)$ which need to be optimized during model training. The CNOT gate connects near-neighbor qubits and generates quantum entanglement to make the model expressive for capturing quantum data correlations.

Typically, repetition of the rotation and entanglement layers increases the expressivity of circuits [46]. In our proposed model, we apply six repetitions of the variational layer to ensure that the model has enough parameters to train, as shown in Fig. 8. Additionally, this repetition increases entanglement layers that enhance the model's expressiveness and improve the ability to capture quantum data correlations.

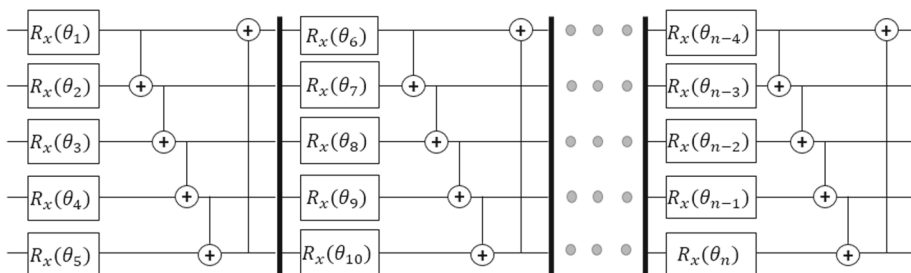
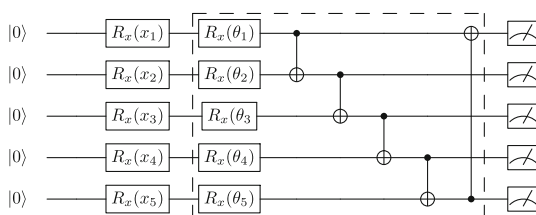


Fig. 8 Repetition of the variational layer utilized in our proposed QRNN model

Fig. 9 A quantum circuit to represent proposed VQC that applied in QRNN cell



5.3 Measurement

Finally, every quantum circuit will be measured using measurement operators to retrieve the classical information. The quantum circuits must be executed many times to get expected values to approximating the deterministic value of a quantum circuit, which is time-consuming. Meanwhile, the Pauli-Z operator mathematically generates the expected value of quantum circuits, providing an efficient technique for measurement [32]. Therefore, QRNN uses the Pauli-Z operator to measure and get the expected value of the VQC.

Specifically, the Pauli-Z operator was chosen for measurement because it is the native and most hardware-efficient measurement basis on current NISQ devices and simulators. Pauli-Z readouts directly correspond to computational base measurements, requiring no additional basis change gates, thereby reducing circuit depth and accumulated noise [13]. If we use any other basis then extra gate should added, which can increase the complexity of model.

Figure 9 represents a proposed VQC for the QRNN cell. The first segment encodes classical information into quantum states, followed by a variational layer, denoted as a dashed line. This variational layer has six repetitions in VQC, as illustrated in Fig. 8. Finally, the measurement operators give the result of the VQC.

Each QRNN cell uses the proposed VQC as its input and hidden layers, while the output layer uses classical methods. Figure 10 presents the overall architecture of the proposed QRNN, constructed by stacking these essential QRNN cells. During training, QRNN employs $VQC(\theta)$ as learnable parameters in place of the weight matrix U and V of RNN, while the output layer utilizes the same weight matrix W , as defined in Eqs. 7 and 8.

$$h_t = f(VQC(x_t, \theta) + VQC(h_{t-1}, \theta)) \quad (7)$$

$$y_t = g(W * h_t) \quad (8)$$

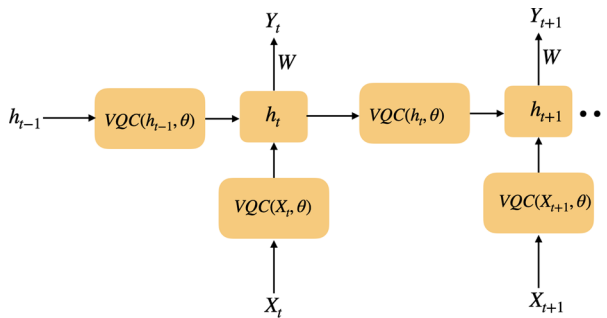


Fig. 10 The architecture of the proposed QRNN model that consists of stacked QRNN cell

Table 2 Statistics of dataset

Social media platform	Dataset name	Sentences	Words
Facebook	English–Hindi	1401	14651
	English–Bengali	782	7243
	English–Telugu	1280	7510
Twitter	English–Hindi	1264	9590
	English–Bengali	464	3370
	English–Telugu	1228	9572
WhatsApp	English–Hindi	1264	9500
	English–Bengali	464	3482
	English–Telugu	998	6499

6 Dataset description

The proposed model implements POS tagging for code-mixed languages, retrieved from the ICON 2016 social media code-mixed dataset in Indian languages.¹ This dataset combines three Indian languages paired with English, such as English–Bengali, English–Hindi, and English–Telugu, collected from three social media platforms: Facebook, WhatsApp, and Twitter. Table 2 provides the statistics of the dataset, including the sentence and word counts of each dataset. Furthermore, Table 3 lists the tag set of the code-mixed dataset and provides the corresponding descriptions.

Code-mixed [47] languages involved two or more languages within speech or text for communication. Generally, the mixing of languages is common in multilingual societies, especially in India, where people frequently change one language to another in their communications. In addition, social media texts also contain code-switching among various languages for reviewing, discussing, and transmitting information. These languages involved multiple source languages that have different grammatical and sentence structures. The processing analysis of code-mixed words poses a significant challenge due to their diverse structure and lexical ambiguity, as various words can have the same meaning.

¹ <http://amitavadas.com/Code-Mixing.html>.

Table 3 Tag set of code-mixed dataset

POS labels	Description
G_N	Noun
G_PRP	Pronoun
G_J	Adjective
G_V	Verb
PSP	Pre- & Postposition
G_R	Adverb
CC	Conjunction
G_PRT	Particles
DT	Determiner
G_SYM	Quantifier
G_X	Residual & Twitter-Specific

7 Experiments and result

QRNN is applied to perform POS tagging in nine different code-mixed datasets, where each dataset represents a unique combination of languages. POS tagging serves as a fundamental sequence labeling task, making it an ideal benchmark to evaluate the effectiveness of the proposed QRNN. Furthermore, in code-mixed data two or more languages appear within the same sentence due to its inherently multilingual structure. Our aim is to assess the versatility and robustness of the model in diverse conditions (Table 3).

Algorithm 1 demonstrates the implementation of POS tagging by the proposed quantum-classical QRNN. Initially, embedding layers are applied to transform text data into word vectors to process the data further. Then, VQCs utilize word vectors to process the data, such as the encoding layer to encode the data into quantum information; the variational layer captures data correlations and maps the data into outcomes. Finally, measurement gives the result of quantum circuits. In the end, classical layers are applied to retrieve each text's tag label.

This section provides a detailed explanation about the experimental setup and ablation studies conducted to evaluate the performance of the proposed model. It includes comparative analysis with classical RNN and state-of-the-art QLSTM, investigation on depths of variational layer, encoding methods, entanglement strategies, and noise models. Finally, these studies demonstrate the efficiency of the proposed model for the sequential labeling task.

7.1 Experimental Setup

All the experiments are implemented on PennyLane framework on its local simulator `default-qubit` except the noise model, which is performed on `default-mixed-simulator`. PennyLane [48] provides a cross-platform based on Python library to implement quantum algorithms. It provides flexibility and a cross-platform environment for QML algorithms, making it a preferable option for our investigation.

Each dataset is divided into 70% for the train, 15% for validation, and 15% for the test to ensure an effective evaluation of the performance of the model. In addition, the details of the hyperparameters utilized are provided in Table 4. The batch size is chosen as 8, because the data size is varied according to language pairs and the size of the dataset is small. The

Algorithm 1 Algorithm to perform POS tagging by proposed QRNN

```

1: Input: Sentence  $S = \{w_1, w_2, \dots, w_n\}$ 
2: Output: POS tag sequence  $T = \{t_1, t_2, \dots, t_n\}$ 

3: Model Parameters:
4:    $\theta_1, \theta_2$  : Trainable parameters of the VQC of input and hidden layer.
5:    $W$  : Trainable parameters of classical output layer.

6: procedure QRNN-TAG( $\theta, W, S$ )
7:   Initialize hidden state  $h_0$ 
8:   for  $w_i$  in  $S$  do
9:      $\mathbf{x}_i \rightarrow \text{Embedding}(\mathbf{w}_i)$  ▷ Classical word embedding
10:     $q_i^{(x)} \leftarrow \text{VQC}(x_i, \theta_1)$  ▷ Quantum encoding + processing of input
11:     $q_i^{(h)} \leftarrow \text{VQC}(h_{i-1}, \theta_2)$  ▷ Quantum processing of previous hidden state
12:     $h_i \leftarrow \tanh(q_i^{(x)}, q_i^{(h)})$  ▷ Quantum-enhanced hidden state
13:     $t_i \leftarrow \text{softmax}(W \cdot h_i)$  ▷ Compute POS tag distribution
14:   end for
15:   return  $\{t_1, t_2, \dots, t_n\}$ 
16: end procedure

```

Table 4 The detail of utilized hyperparameter in the experiments

Hyperparameter	Value
Batch Size	8
Embedding Dimension	32
Hidden Dimension	32
Learning Rate	0.01
Optimizer	SGD
Loss Function	Cross-entropy Loss

embedding and hidden dimensions are set to 32 to provide a balance between efficiency and complexity of the model. The learning rate is set to be 0.01 to ensure stable learning. Further, to improve generalization and prevent overfitting, we add dropout at the hidden and input layer, label smoothing, and early stop to avoid overfitting.

Currently, most QML algorithms use classical optimization techniques to train the models. In our experiments, stochastic gradient optimization is used to optimize the VQC parameters and weight matrix W . It is a fundamental optimizer tool for machine learning and neural network models, which helps to escape the model from local minima and provides a wider range loss landscape. Next, selecting an appropriate loss function is a crucial aspect of the learning process of any machine learning model. POs tagging is a multi-classification task, while negative log likelihood is a well-suited option for multi-classification tasks. Therefore, we opted for the negative logarithmic likelihood as a loss function. Finally, accuracy is selected as the performance matrix of the proposed model.

7.2 Comparison with classical RNN baseline

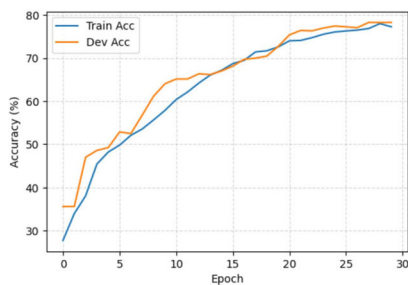
To evaluate the advantage of the proposed model, we compared the proposed model with the classical baseline RNN. Both models are trained in identical experimental settings, such as the same data spits, learning rate, and hyperparameter as mentioned in Table 3 to make

Table 5 Required number of parameters for RNN and QRNN

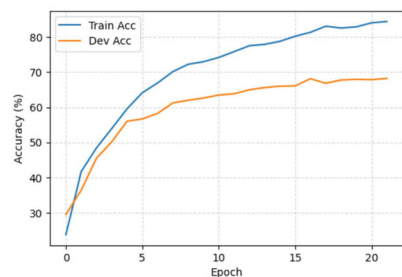
	Number of parameters
Classical RNN	256
QRNN	126

Table 6 Comparison between the accuracy of the proposed QRNN model and RNN to perform POS tagging

Source	Language	Accuracy of QRNN	Accuracy of RNN
Facebook	English–Hindi	73.56	70.84
	English–Bengali	77.38	70.34
	English–Telugu	66.14	65.50
Twitter	English–Hindi	76.32	73.98
	English–Bengali	67.51	62.62
	English–Telugu	66.41	61.51
WhatsApp	English–Hindi	65.10	72.60
	English–Bengali	76.0	66.74
	English–Telugu	60.24	63.47



(a) Training and validation accuracy curves for QRNN output



(b) Training and validation accuracy curves for Classical RNN output

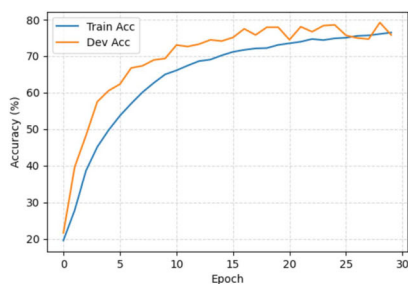
Fig. 11 Comparison between classical RNN and proposed QRNN model

a fair evaluation. The comparison between models based on various aspects such as overall classification accuracy, total number of trainable parameters, and learning capability allows us to evaluate the capability of QRNN to learn sequence labeling compared to classical RNN.

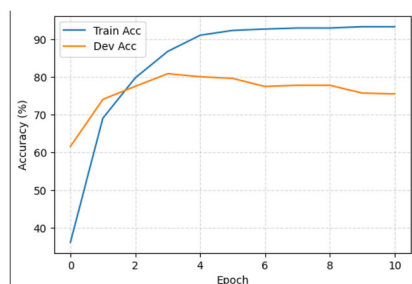
The QRNN requires significantly fewer trainable parameters than the classical RNN in this experiment. Table 5 shows the required number of both models that show that QRNN achieves superior performance while using a substantially smaller number of parameters. Table 6 presents the performance comparison between the proposed QRNN and the classical RNN in all datasets. The QRNN consistently outperforms the classical RNN in every dataset except the WhatsApp English–Telugu dataset. Furthermore, the graph illustrated in Fig. 11 mentioned the trends of the dataset such as the Facebook English–Telugu dataset. It highlights the difference in generalization behavior between the two models. The classical RNN exhibits signs of overfitting, with a large gap between training and validation performance, whereas the QRNN demonstrates more stable and generalized learning across epochs.

Table 7 Comparison between the accuracy of the amplitude encoding and angle encoding

Source	Language	Amplitude encoding	Angle encoding
Facebook	English–Hindi	73.43	73.56
	English–Bengali	68.0	77.38
	English–Telugu	65.62	66.14
Twitter	English–Hindi	73.50	76.32
	English–Bengali	65.36	67.51
	English–Telugu	52.29	66.41
WhatsApp	English–Hindi	65.85	65.10
	English–Bengali	73.32	76.0
	English–Telugu	60.41	60.24



(a) Training and validation accuracy curves for angle encoding



(b) Training and validation accuracy curves for amplitude encoding

Fig. 12 Comparison between angle encoding and amplitude encoding

7.3 Effect of data encoding methods

We experimented with two widely used data encoding methods, such as angle encoding and amplitude encoding, to compare the effect of data encoding methods. However, the proposed model utilizes angle encoding, which is an efficient data encoding method for QRNNs. The amplitude encoding is also used to check the learning ability of the sequences using the QRNN model. The evaluation of both encoding models shows the impact of data representation and the overall accuracy of the proposed model.

Table 7 presents the comparison of the results between angle encoding and amplitude encoding. Overall, angle encoding shows higher accuracy or comparable accuracy compared to amplitude encoding. In particular, Fig. 12 shows training trends in the Facebook English–Hindi dataset that shows more generalized QRNN behaviors with angle encoding compared to amplitude encoding.

7.4 Effect of entanglement strategies

The variational layer can be classified into various types according to its different entanglement strategies such as nearest-neighbor (NN), circuit-block (CB) and all-to-all (AA) as illustrated in Figs. 13 and 14. Each qubit is entangled only with its immediate neighbor, resulting in minimal connectivity and limited expressive power. The AA topology, on the

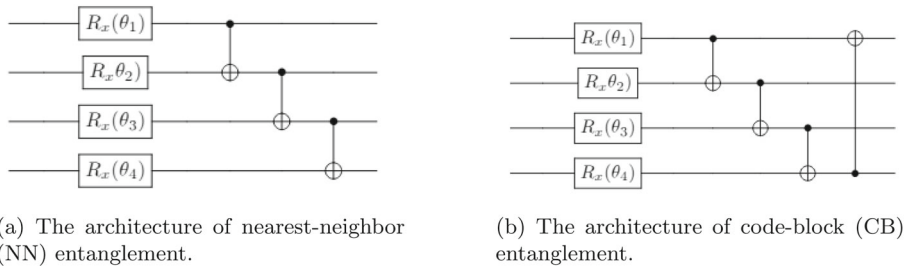


Fig. 13 The architecture of various entanglement approaches

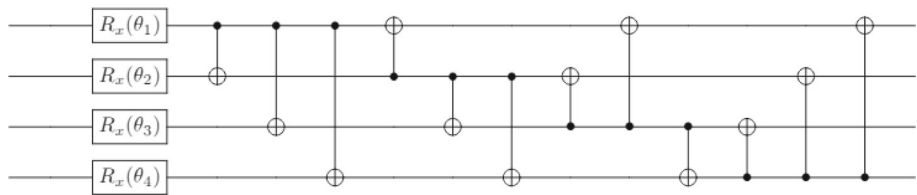


Fig. 14 The architecture of all-to-all (AA) entanglement

Table 8 Comparison between the accuracy of the NN, CB, and AA

Source	Language	NN	CB	AA
Facebook	English–Hindi	66.69	73.56	69.02
	English–Bengali	70.32	77.38	66.36
	English–Telugu	61.45	66.14	63.15
Twitter	English–Hindi	69.3	76.32	73.0
	English–Bengali	61.24	67.51	68.4
	English–Telugu	65.41	66.41	68.32
WhatsApp	English–Hindi	61.23	65.10	64.9
	English–Bengali	66.76	76.0	78.23
	English–Telugu	56.55	60.24	61.43

other hand, entangles every qubit with all others, providing strong global correlations but significantly increasing circuit depth and computational complexity. The CB topology offers a balanced alternative by forming a circular ring structure in which each qubit interacts with multiple neighboring qubits. Therefore, QRNN utilizes CB as a variant layer.

Entanglement in variational layer contributes significantly in expressive power of VQCs. To investigate the effect of the different entanglement approaches in the performance of the proposed model, we utilize three different entanglement approaches such as AA, NN, and CB. However, the proposed model utilizes the entanglement of CBs in the variational layer. Table 8 shows the comparative results, where the CB entanglement outperforms the other strategies. Figure 15 shows a comparison between Facebook English–Bengali.

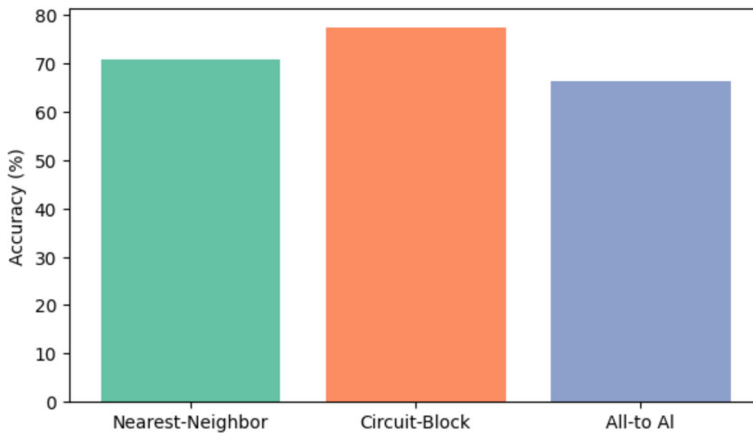


Fig. 15 The result comparison analysis of different entanglement strategies

Table 9 Comparison between the accuracy of various depths of variational layer

Source	Language	2	4	6
Facebook	English–Hindi	60.01	67.61	73.56
	English–Bengali	68.0	52.1	77.38
	English–Telugu	61.0	53.9	66.14
Twitter	English–Hindi	68.5	71.3	76.32
	English–Bengali	60.24	63.4	67.51
	English–Telugu	48.73	59.3	66.41
WhatsApp	English–Hindi	61.10	62.23	65.10
	English–Bengali	66.76	70.45	76.0
	English–Telugu	54.06	55.6	60.24

7.5 Effect of variational layer depth

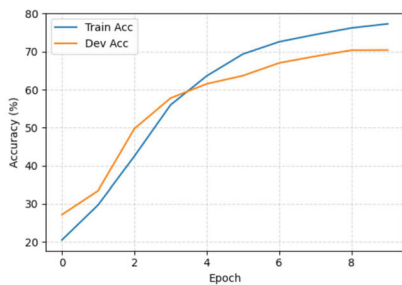
To investigate the effect of depth of the variational layer on model performance, we performed experiments with various numbers of variational layer $L \in \{2, 4, 6\}$, where each variational layer includes additional trainable parameters and gate operations. The deeper circuits can handle more complex correlations and transformations, potentially enabling the model to learn richer sequential patterns. However, it comes at the cost of increased computational overhead and challenging optimization landscapes. Table 9 shows the accuracy of each layer. Overall, the number of variational layer 6 gives better accuracy.

7.6 Performance of QRNN on noisy model

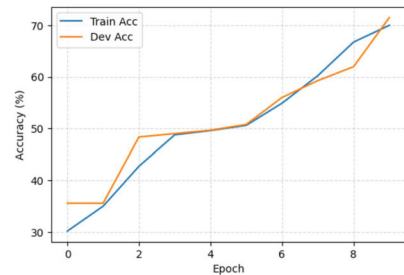
The evaluation of the robustness of the proposed model on real quantum hardware is very crucial for its practical applicability. However, executing large-scale experiments on real devices is costly and limited to accessing real quantum devices, and therefore, most of the experiments are run on a noisy simulator. In this study, we simulate QRNN on a noisy PennyLane's `default.mixed` simulator, which provides real quantum hardware behavior. This noisy model adds errors such as gate imperfections and measurement inaccuracies. To

Table 10 Comparison between the accuracy of default.mixed and default.qubit

Source	Language	default.mixed	default.qubit
Facebook	English–Hindi	71.69	68.89
	English–Bengali	68.0	68.77
	English–Telugu	61.0	53.9
Twitter	English–Hindi	68.5	68.64
	English–Bengali	60.24	47.79
	English–Telugu	59.81	50.73
WhatsApp	English–Hindi	65.10	49.93
	English–Bengali	66.76	59.45
	English–Telugu	54.06	55.66



(a) Training and validation accuracy curves for noisy model



(b) Output of simulator

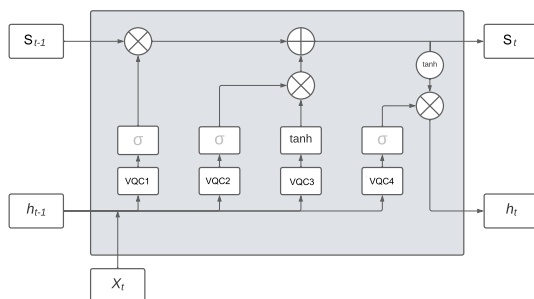
Fig. 16 Training and validation accuracy curves for simulator

evaluate the model, we run QRNN on noisy model and default.qubit to ten epochs. By analyzing the results summarized in Table 10, we observe that the QRNN model shows a comparative result on the noisy also that shows robustness the proposed QRNN. Fig 16 shows a clear trend of QRNN learning both the noisy model and the simulator for a 10 epoch.

7.7 Comparison with QLSTM

QLSTM [21] is a state-of-the-art quantum-enhanced LSTM model, in which each gate of the classical LSTM is replaced by VQC as shown in Fig. 17. It uses angle encoding as data encoding, CB as variational layer QLSTM, and a single variational layer in its VQC. QLSTM is applied to perform POS tagging on a mixed code dataset to evaluate its effectiveness. During experiments, each dataset is divided into training and testing subsets, with a distribution ratio of 80% for training and 20% for testing. For comparison of QRNN with QLSTM, we also split the dataset with a distribution ratio of 80% for training and 20% only for this experiment.

This table shows a comparison between QLSTM and QRNN precision, where QRNN performs better for all datasets except the dataset, except for the English–Telugu dataset. The present QRNN model represents a notable improvement over the previous state-of-the-art QLSTM model in the performing of sequence labeling. We observe that QLSTM uses a single variational layer in its VQCs, while QRNN uses six repetitions of variational layers.

Fig. 17 An diagram of a quantum long short-term memory [21]**Table 11** Comparison between the accuracy of the proposed QRNN model and QLSTM [21] to perform POS labeling for the code-mixed dataset

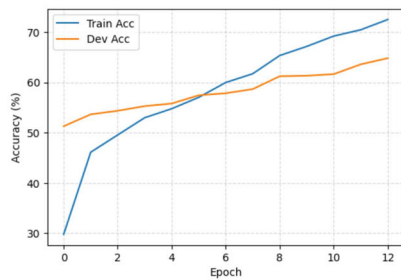
Source	Language	Accuracy of proposed QRNN	Accuracy of QLSTM
Facebook	English–Hindi	71.69	56.9
	English–Bengali	68.0	52.1
	English–Telugu	61.0	53.9
Twitter	English–Hindi	68.5	49.94
	English–Bengali	60.24	55.25
	English–Telugu	48.73	45.38
WhatsApp	English–Hindi	65.10	49.93
	English–Bengali	66.76	59.45
	English–Telugu	54.06	55.66

This QRNN design contains more tunable parameters and entanglement layers that enhance the expressiveness of quantum circuits and model learning capabilities. This novel model shows the potential of QRNN for efficiently labeling input sequences.

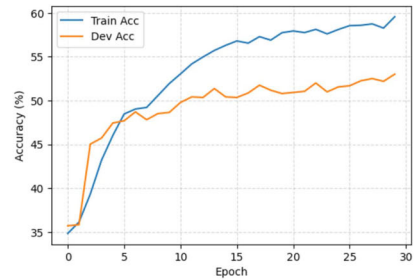
8 Discussion

The experimental results show that QRNN achieves superior performance compared to QLSTM and classical RNN with their corresponding experimental setup. QRNN required fewer parameters compared to RNN, which shows its ability to learn. In addition, a small gap between training and validation accuracy as shown in Fig. 11 represents its generalization capability compared to its classical RNN. Moreover, results of QRNN on noisy model give comparative analysis compared to simulator outcome that indicates robustness of proposed model. These results confirm the potential of QRNN architectures for handling sequential labeling tasks.

In the data encoding context, the performance of QRNN with angle encoding achieves higher accuracy and a more generalized result. The various approaches to entanglement start-ups also impact the performance of QRNN. For some datasets including Twitter English–Bengali, WhatsApp English–Telugu, and Twitter English–Telugu, the AA entanglement has a higher precision compared to the CB entanglement states. However, AA entanglement can increase the complexity of the model. Overall, CB entanglement is a good fit for QRNN.



(a) Training and validation accuracy curves for WhatsApp English-Telugu



(b) Training and validation accuracy curves for Twitter English-Telugu

Fig. 18 This graph shows limitation of QRNN

However, we observed that QRNN does not perform well in WhatsApp Telugu and Twitter Telugu, as shown in Fig. 18. This graph shows how QRNN is not well generalized for these datasets. In contrast, QRNN shows higher accuracy on Facebook dataset compared to WhatsApp and Telugu dataset. As Tables 6 and 11 show, classical RNN and QLSTM give a better result compared to QRNN for the WhatsApp Telugu dataset. The average length of the Facebook dataset is higher than that of the WhatsApp and Twitter dataset. One possible explanation for this behavior is that QRNN benefits from longer sequences data contexts. The possible limitations of our model are the reduced performance on shorter or less structured text sequences. Therefore, to overcome this limitation, our future work will check the performance of QRNN on various length datasets.

9 Conclusion

This paper introduces a hybrid quantum-classical QRNN that utilizes a quantum circuit model and replaces RNN cells with VQCs. We aim to present the QRNN model as simple and efficient to demonstrate its applicability in NLP applications, especially in sequence labeling. The proposed model applies to perform POS tagging on code-mixed datasets. The experimental results show the improved performance of QRNN compared to the previous QLSTM model for POS tagging because of its repetition of variational layers, which enhances the model's capability. In addition, we compare QRNN with traditional baseline RNN models. The experimental data show that QRNN has better learning ability and generalization compared to its classical counterpart. Furthermore, this study presents a comprehensive ablation study to analyze the impact of different data encoding methods, entanglement strategies, and the number of variational layers.

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Author Contributions Shyambabu Pandey contributed to implementation and writing. Partha Pakray contributed to idea generation, writing, and supervising. Fabio Massimo Zanzotto contributed to idea generation, writing, and supervising.

Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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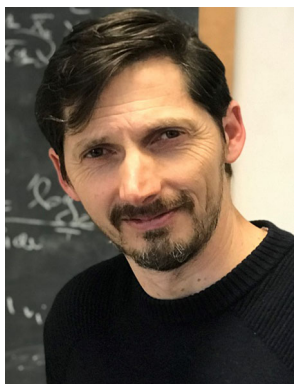


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